25900270 **Reg.No : .....**

Name : .....

MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS) Regular EXAMINATION October 2025

Third SEMESTER

Discipline Specific Core Course (DSC) - MG3DSCBTG201, MG3DSCBBT201

CELL BIOLOGY AND GENETICS

(2024 ADMISSION ONWARDS)

Duration: 1 Hour 30 Minutes **Maximum Marks: 50**

Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(I) and Appreciation(Ap)

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A

One word or sentence

Answer any **10** questions

Each question carries **2** marks

1. What are the different types of Sex linked Inheritance?. [U] / [CO3]
2. What is macroautophagy? [U] / [CO2]
3. Define Mutation. [U] / [CO3]
4. Name two functions of cholesterol in the plasma membrane. [U] / [CO1]
5. Name the scientist who proposed the inheritance of A and B blood groups?. [K] / [CO3]
6. Name the enzyme which is present in peroxisomes to break down hydrogen peroxide. [K] / [CO2]
7. Name the organelle which plays a key role in forming primary lysosomes. [K] / [CO2]
8. What is a karyotype? [K] / [CO3]
9. What does non-random mating cause in a population? [K] / [CO3]

10. Which type of transport involves the movement of substances from high to low concentration without energy? [K] / [CO1]

11. Which type of Sex determination system is present in Drosophila Melanogaster? [K] / [CO3]

12. What is euchromatin? [U] / [CO2]

[2x10 = 20]

Part B

Short Essay

Answer any **5** questions

Each question carries **4** marks

13. Mention the characteristics of prokaryotic and eukaryotic cells [U] / [CO1]

14. Differentiate sex-linked traits and autosomal trait. [U] / [CO3]

15. Write a note on the structure and functions of glycocalyx. [U] / [CO1]

16. Discuss the diverse roles played by integrins in ECM. [K] / [CO1]

17. Explain why mitochondria is called the powerhouse of the cell. [U] / [CO2]

18. Describe the different stages of crossing over. [U] / [CO3]

19. Briefly explain Mendalian laws. [U] / [CO3]

[4x5 = 20]

Part C

Essay type questions

Answer any **1** questions

Each question carries **10** marks

20. Describe the structure and functions of endoplasmic reticulum in eukaryotic cells. [U] / [CO2]

21. Describe in detail the extra-chromosomal inheritance shown by mitochondria with suitable examples. [U] / [CO3]

[10x1 = 10]